

SNOWFLAKE PRACTICE EXERCISES – JSON & XML

EXERCISE 1 – JSON NORMAL (employees.json)

Dataset (10 records):

```
[
  {"id":101,"name":"John Doe","department":"HR","salary":5000},
  {"id":102,"name":"Jane Smith","department":"Finance","salary":6000},
  {"id":103,"name":"Michael Brown","department":"IT","salary":5500},
  {"id":104,"name":"Linda White","department":"Marketing","salary":5200},
  {"id":105,"name":"James Black","department":"IT","salary":5800},
  {"id":106,"name":"Emma Green","department":"HR","salary":5100},
  {"id":107,"name":"Robert King","department":"Finance","salary":6200},
  {"id":108,"name":"Olivia Scott","department":"Marketing","salary":5300},
  {"id":109,"name":"William Lee","department":"IT","salary":5600},
  {"id":110,"name":"Sophia Turner","department":"HR","salary":5000}
]
```

Tasks:

1. Create a table employees_json with a VARIANT column raw.
2. Upload JSON file to a Snowflake stage.
3. Load the JSON file into the table using COPY INTO.
4. Query employee id, name, and department from the JSON column.

EXERCISE 2 – JSON NESTED (projects.json)

Dataset (10 records):

```
[
  {"id":101,"name":"John Doe","projects":[{"project_name":"Website Redesign","duration":3}, {"project_name":"Employee Portal","duration":4}]},
  {"id":102,"name":"Jane Smith","projects":[{"project_name":"Financial Report","duration":5}, {"project_name":"Audit Compliance","duration":2}]},
  {"id":103,"name":"Michael Brown","projects":[{"project_name":"IT Security","duration":6}, {"project_name":"Cloud Migration","duration":8}]},
  {"id":104,"name":"Linda White","projects":[{"project_name":"Marketing Campaign","duration":3}, {"project_name":"Social Media Outreach","duration":2}]},
  {"id":105,"name":"James Black","projects":[{"project_name":"Mobile App Dev","duration":4}, {"project_name":"Server Optimization","duration":5}]},
  {"id":106,"name":"Emma Green","projects":[{"project_name":"HR Portal","duration":3}, {"project_name":"Employee Training","duration":2}]},
  {"id":107,"name":"Robert King","projects":[{"project_name":"Financial Audit","duration":4}, {"project_name":"Budget Planning","duration":3}]},
  {"id":108,"name":"Olivia Scott","projects":[{"project_name":"Marketing Analytics","duration":3}, {"project_name":"Campaign Review","duration":2}]},
  {"id":109,"name":"William Lee","projects":[{"project_name":"Server Maintenance","duration":5}, {"project_name":"Network Upgrade","duration":4}]},
  {"id":110,"name":"Sophia Turner","projects":[{"project_name":"Employee Onboarding","duration":2}, {"project_name":"Benefits Portal","duration":3}]}
]
```

Tasks:

1. Create a table employee_projects_json with VARIANT column raw.
2. Upload and load JSON file into the table.
3. Flatten projects array and display employee id, name, project_name, and duration.

EXERCISE 3 – XML NORMAL (employees.xml)

Dataset (10 records):

```

<employees>
  <employee id="101"><name>John
Doe</name><department>HR</department><salary>5000</salary></employee>
  <employee id="102"><name>Jane
Smith</name><department>Finance</department><salary>6000</salary></employee>
  <employee id="103"><name>Michael
Brown</name><department>IT</department><salary>5500</salary></employee>
  <employee id="104"><name>Linda
White</name><department>Marketing</department><salary>5200</salary></employee>
  <employee id="105"><name>James
Black</name><department>IT</department><salary>5800</salary></employee>
  <employee id="106"><name>Emma
Green</name><department>HR</department><salary>5100</salary></employee>
  <employee id="107"><name>Robert
King</name><department>Finance</department><salary>6200</salary></employee>
  <employee id="108"><name>Olivia
Scott</name><department>Marketing</department><salary>5300</salary></employee>
  <employee id="109"><name>William
Lee</name><department>IT</department><salary>5600</salary></employee>
  <employee id="110"><name>Sophia
Turner</name><department>HR</department><salary>5000</salary></employee>
</employees>

```

Tasks:

1. Create table employees_xml with VARIANT column raw.
2. Upload XML file to a stage.
3. Load XML into the table using COPY INTO.
4. Query employee id, name, and department.

----- EXERCISE 4 – XML NESTED (projects.xml) -----

Dataset (10 records):

```

<employees>
  <employee id="101"><name>John Doe</name><projects>
    <project><project_name>Website
Redesign</project_name><duration>3</duration></project>
    <project><project_name>Employee
Portal</project_name><duration>4</duration></project>
  </projects></employee>
  <employee id="102"><name>Jane Smith</name><projects>
    <project><project_name>Financial
Report</project_name><duration>5</duration></project>
    <project><project_name>Audit
Compliance</project_name><duration>2</duration></project>
  </projects></employee>
  <employee id="103"><name>Michael Brown</name><projects>
    <project><project_name>IT
Security</project_name><duration>6</duration></project>
    <project><project_name>Cloud
Migration</project_name><duration>8</duration></project>
  </projects></employee>
  <employee id="104"><name>Linda White</name><projects>
    <project><project_name>Marketing
Campaign</project_name><duration>3</duration></project>
    <project><project_name>Social Media
Outreach</project_name><duration>2</duration></project>
  </projects></employee>
  <employee id="105"><name>James Black</name><projects>
    <project><project_name>Mobile App
Dev</project_name><duration>4</duration></project>
    <project><project_name>Server

```

```

Optimization</project_name><duration>5</duration></project>
  </projects></employee>
  <employee id="106"><name>Emma Green</name><projects>
    <project><project_name>HR
Portal</project_name><duration>3</duration></project>
    <project><project_name>Employee
Training</project_name><duration>2</duration></project>
  </projects></employee>
  <employee id="107"><name>Robert King</name><projects>
    <project><project_name>Financial
Audit</project_name><duration>4</duration></project>
    <project><project_name>Budget
Planning</project_name><duration>3</duration></project>
  </projects></employee>
  <employee id="108"><name>Olivia Scott</name><projects>
    <project><project_name>Marketing
Analytics</project_name><duration>3</duration></project>
    <project><project_name>Campaign
Review</project_name><duration>2</duration></project>
  </projects></employee>
  <employee id="109"><name>William Lee</name><projects>
    <project><project_name>Server
Maintenance</project_name><duration>5</duration></project>
    <project><project_name>Network
Upgrade</project_name><duration>4</duration></project>
  </projects></employee>
  <employee id="110"><name>Sophia Turner</name><projects>
    <project><project_name>Employee
Onboarding</project_name><duration>2</duration></project>
    <project><project_name>Benefits
Portal</project_name><duration>3</duration></project>
  </projects></employee>
</employees>

```

Tasks:

1. Create table employee_projects_xml with VARIANT column raw.
2. Upload XML file to a stage.
3. Load XML into the table using COPY INTO.
4. Flatten projects array and display employee id, name, project_name, and duration.